

Topological Projections of 7D Toroidal Manifolds into Minkowski Spacetime: Lissajous Knot Dynamics and the 48-Vertex Matrix of Elementary Particles

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Abstract

The Standard Model traditionally treats quarks and gauge bosons as zero-dimensional point particles with intrinsic quantum numbers. This paper proposes a geometric paradigm shift via the Heptagonal Unitary Field Theory (HUFT), wherein fundamental particles are modeled as topological intersection artifacts—specifically Lissajous knots—resulting from the continuous projection of a 7-dimensional toroidal manifold (T^7) onto 4-dimensional Minkowski spacetime ($\mathcal{M}^4$). By mapping a 288-point double-toroidal cut, we derive a fundamental 48-vertex sub-matrix that perfectly bounds the observed degrees of freedom for quarks and leptons. Furthermore, we explore the topology of W and Z bosons, as well as microscopic black holes, as tied flux tubes (knots) within this T^7 fabric, and postulate that physical constants such as π act as state-dependent functions within T^7 , stabilizing as constants only upon projection into $\mathcal{M}^4$.

1 Introduction

Current discrepancies in quantum field theory, particularly regarding gravity and the arbitrary parameters of the Standard Model, suggest that 4D Minkowski space ($\mathcal{M}^4$) is a limited projection of a higher-dimensional geometry. Building upon topological geons and the treatment of particles as knotted flux tubes, we introduce a framework where the fabric of spacetime is a continuous, grazing projection from a 7D manifold, denoted as T^7 .

2 The Projection Operator and Lissajous Trajectories

Let the higher-dimensional space be modeled as a highly coherent 7D torus T^7 . The observable universe exists as a discrete 4D slice, bounded by Planck time intervals. We define the projection operator Π mapping the continuous 7D states to our observable $\mathcal{M}^4$:

$$\Pi_{\mathcal{M}^4} : T^7 \rightarrow \mathcal{M}^4 \tag{1}$$

Particles, specifically quarks, are not localized points but rather the rolling, intersecting trajectories of 7D Lissajous shapes grazing the $\mathcal{M}^4$ boundary. If we parameterize the rolling trajectory by a variable τ , the observed quark is the localized knot intersection.

3 The 288-Point Double Toroid and 48-Vertex Matrix

To discretize the continuous projection for quantum observation, we apply a double-toroidal cut, yielding a crystalline matrix of 288 intersection points. These points represent the absolute phase-space boundaries of the projection.

Applying spatial and dimensional symmetries, this 288-point geometry reduces to a core matrix of 48 vertices:

$$V_{\text{core}} = \frac{V_{288}}{6} = 48 \quad (2)$$

This 48-vertex structure maps exactly to the foundational degrees of freedom in the Standard Model. Considering the 12 fundamental fermions (6 quarks, 6 leptons) and their 12 antiparticles (totaling 24), and multiplying by their chiral/spin states, the 48 vertices provide a purely geometric origin for particle generation and color charge distribution, removing the need for arbitrary intrinsic properties.

4 Knotted Topologies: W/Z Bosons and Black Holes

Within this topological framework, mass and charge are derived not from intrinsic scalar fields, but from the complexity of the knotting in the T^7 manifold. While quarks represent continuous Lissajous grazings with low topological crossing numbers, the weak force mediators (W and Z bosons) represent specific, highly knotted flux configurations.

Let K be a knot in the T^7 manifold. The mass m of the resulting particle in $\mathcal{M}^4$ is mathematically proportional to the knot crossing number $c(K)$ and the topological winding density ω :

$$m_{W,Z} \propto c(K) \int_{T^7} \omega \wedge d\omega \quad (3)$$

These tight topological ties require immense energy to unbind, manifesting as massive particles with short topological stability (half-lives) before decaying into simpler Lissajous trajectories (fermions).

Similarly, microscopic black holes are modeled not as gravitational singularities of infinite density, but as macroscopic topological knots in the $T^7 \rightarrow \mathcal{M}^4$ projection. In regions of extreme energy density, the Lissajous trajectories become so tightly wound that the projection operator $\Pi_{\mathcal{M}^4}$ becomes degenerate. The local curvature of $\mathcal{M}^4$ folds back upon itself, creating a bounded region where the determinant of the induced metric vanishes, effectively trapping the local spacetime within a closed topological loop.

5 Dimensional Variance of Physical Constants

A critical consequence of the T^7 framework is the reevaluation of fundamental constants. We postulate that the constant π , conventionally fixed in $\mathcal{M}^4$, is fundamentally a state-dependent function within the higher manifold:

$$\pi_{T^7} = f(\tau) \tag{4}$$

Multiple parallel projections sliding through T^7 in discrete Planck pulses may experience varying functional values of π_{T^7} . It is only the geometric constraints of the $\mathcal{M}^4$ boundary that force the collapse of this function into the strict constant $\pi \approx 3.14159$.

6 Conclusion

The HUFT framework, by treating the observable universe as a 48-vertex topological footprint of a 288-point rolling T^7 geometry, offers a purely geometric explanation for the Standard Model. By redefining particles as Lissajous knots, weak bosons as high-density topological crossings, and constants as localized functional collapses, we pave the way for a mathematically rigorous unification of quantum topology and relativity.

References

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